

JavaScript Module 1 - Chapter 1: Introduction to JavaScript

What is JavaScript?

JavaScript is a high-level, interpreted, dynamically typed, multi-paradigm programming language used to make web pages interactive and dynamic. It runs inside web browsers and can also run on servers using Node.js.

Programming Language

A programming language is a way for humans to write instructions that computers can execute. Computers understand machine code (0s and 1s), so programming languages make development easier.

Role of HTML, CSS and JavaScript

HTML provides the structure of a webpage, CSS provides styling, and JavaScript provides behavior and interactivity such as button clicks, form validation, animations, and dynamic content.

Where JavaScript Runs

- Web Browsers (Chrome, Firefox, Edge, Safari)
- Server-side using Node.js

Main Features

- High-level language with automatic memory management.
- Interpreted/JIT executed by JavaScript engines.
- Dynamically typed variables.
- Supports object-oriented, functional and procedural programming.
- Event-driven programming model.
- Cross-platform support.

Simple Examples

```
console.log("Hello JavaScript");
```

```
let x = 10;  
x = "Hello";
```

Interview Questions

1. What is JavaScript?
2. Difference between Java and JavaScript?
3. Where can JavaScript run?
4. Name five features of JavaScript.

Practice

Explain the relationship between HTML, CSS, and JavaScript in your own words. Run basic `console.log` statements in the browser console.